Lab Report

Course Title: Computer Networks Laboratory
Course Code: CSE-3634

Spring-2023

Lab No: 10

Name of Labwork: Implement OSPF dynamic routing protocol on the network, Created in previous lab work (Lab9).

TEAM_DIVERGENT

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1. Introduction:

This lab report presents the findings and insights gained from a comprehensive project undertaken to demonstrate proficiency in network design and routing using the GNS3, a network simulation software. In lab work 10, we extend our exploration of network dynamics, building upon the foundation established in our previous endeavor, Lab Work 9. In the previous lab, we meticulously partitioned the IP address range 192.168.111.0/24 into four distinct subnets and employed static routing protocols to establish network connectivity. In this subsequent phase, we introduce the OSPF (Open Shortest Path First) dynamic routing protocol into our network architecture while leveraging the insights provided by Wireshark, a powerful network protocol analyzer. This transition from static to dynamic routing, complemented by Wireshark's capabilities, seeks to enhance network efficiency and adaptability. It enables the network to autonomously respond to changes in topology and traffic demands while providing invaluable insights into packet-level communication, a crucial aspect of modern network design and management.

2. Description:

This lab experiment entails the implementation of the **OSPF** *(Open Shortest Path First)* dynamic routing protocol into the existing network infrastructure established in Lab Work 9. This phase of the project aims to transition from static routing to dynamic routing, enhancing network efficiency, scalability, and adaptability. The practical tasks include configuring OSPF on Cisco routers, enabling interfaces, defining area assignments, and optionally assigning router IDs. This lab work builds upon the prior subnetting and static routing work, leveraging OSPF's features, such as support for variable-length subnet masks (VLSMs), multicast communication, rapid convergence, load balancing, security features, and the ability to tag external routes. It serves as a pivotal step in optimizing network performance and routing control within the network topology.

3. Routing Table:

Destination Subnet	Net id	First ID	Last ID	Broadcast	Subnet Mask
192.168.111.0/26	192.168.111.0	192.168.111.1	192.168.111.62	192.168.111.63	255.255.255.192
192.168.111.64/26	192.168.111.64	192.168.111.65	192.168.111.126	192.168.111.127	255.255.255.192
192.168.111.128/26	192.168.111.128	192.168.111.129	192.168.111.190	192.168.111.191	255.255.255.192
192.168.111.192/26	192.168.111.192	192.168.111.193	192.168.111.254	192.168.111.255	255.255.255.192

4. Configuring IP, Subnet Mask and Gateway of the VPCs:

PC1

```
PC1> ip 192.168.111.30 255.255.255.192 192.168.111.1
Checking for duplicate address...
PC1 : 192.168.111.30 255.255.255.192 gateway 192.168.111.1

PC1> save
Saving startup configuration to startup.vpc
.  done
```

PC2

```
PC2> ip 192.168.111.80 255.255.255.192 192.168.111.65
Checking for duplicate address...
PC1 : 192.168.111.80 255.255.255.192 gateway 192.168.111.65

PC2> save
Saving startup configuration to startup.vpc
.  done
```

PC3

```
PC3> ip 192.168.111.150 255.255.255.192 192.168.111.129
Checking for duplicate address...
PC1 : 192.168.111.150 255.255.255.192 gateway 192.168.111.129

PC3> save
Saving startup configuration to startup.vpc
.  done
```

PC4

```
PC4> ip 192.168.111.200 255.255.255.192 192.168.111.193
Checking for duplicate address...
PC1 : 192.168.111.200 255.255.255.192 gateway 192.168.111.193

PC4> save
Saving startup configuration to startup.vpc
.  done
```

5. OSPF Configuration of the Routers:

Figure 1: OSPF of router 1

Figure 2: OSPF of router 2

Figure 3: OSPF of router 3

Figure 4: OSPF of router 4

6. Ping Connectivity:

To verify the connectivity and assess the network link's functionality, a ping test is conducted between all the devices.

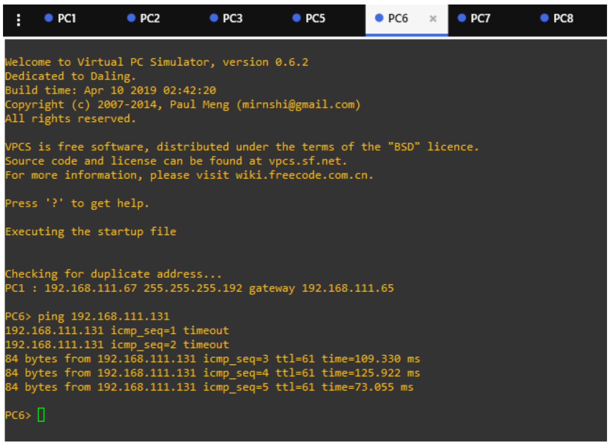

Figure 5: Ping check between PC6 to PC4

10. Result Analysis:

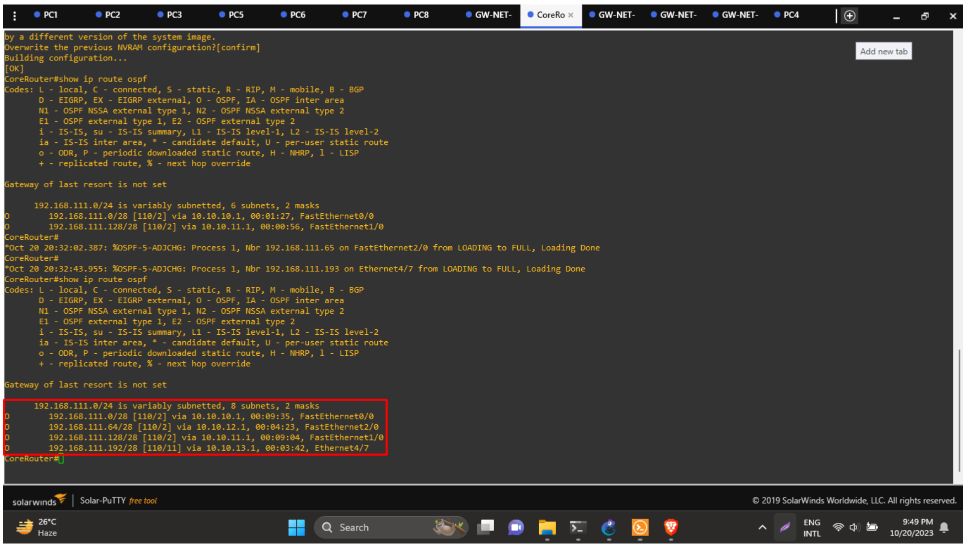

Figure 6: Core OSPF

The implementation of the OSPF (Open Shortest Path First) dynamic routing protocol in our network, as observed through various metrics and configurations, yielded several noteworthy outcomes:

Routing Tables: One of the most evident changes was the transformation of routing tables. OSPF dynamically updated routing entries as routers exchanged Link State Advertisements (LSAs), reflecting the current network topology. This dynamic routing approach resulted in more accurate and efficient routing decisions.

Network Connectivity: Compared to the previous use of static routing, OSPF significantly improved network connectivity. It efficiently adapted to changes in the network, ensuring that data packets reached their destinations with minimal delay. This was particularly evident when dealing with evolving network conditions, such as link failures or the addition of new subnets.

Network Stability: OSPF's rapid convergence mechanisms were apparent in the network's stability. When faced with link failures or other topology changes, OSPF quickly recalculated routes, minimizing downtime and ensuring seamless communication between subnets.

11. Conclusion:

In this lab, we affirmed the value of OSPF as a dynamic routing protocol in modern network design. Its ability to adapt to changing network conditions, optimize routing, and enhance network stability, along with its security features, positions OSPF as a robust and versatile choice for large enterprise networks. This transition represents a pivotal step in our network's evolution, empowering it to meet the demands of a dynamic and ever-evolving digital landscape.